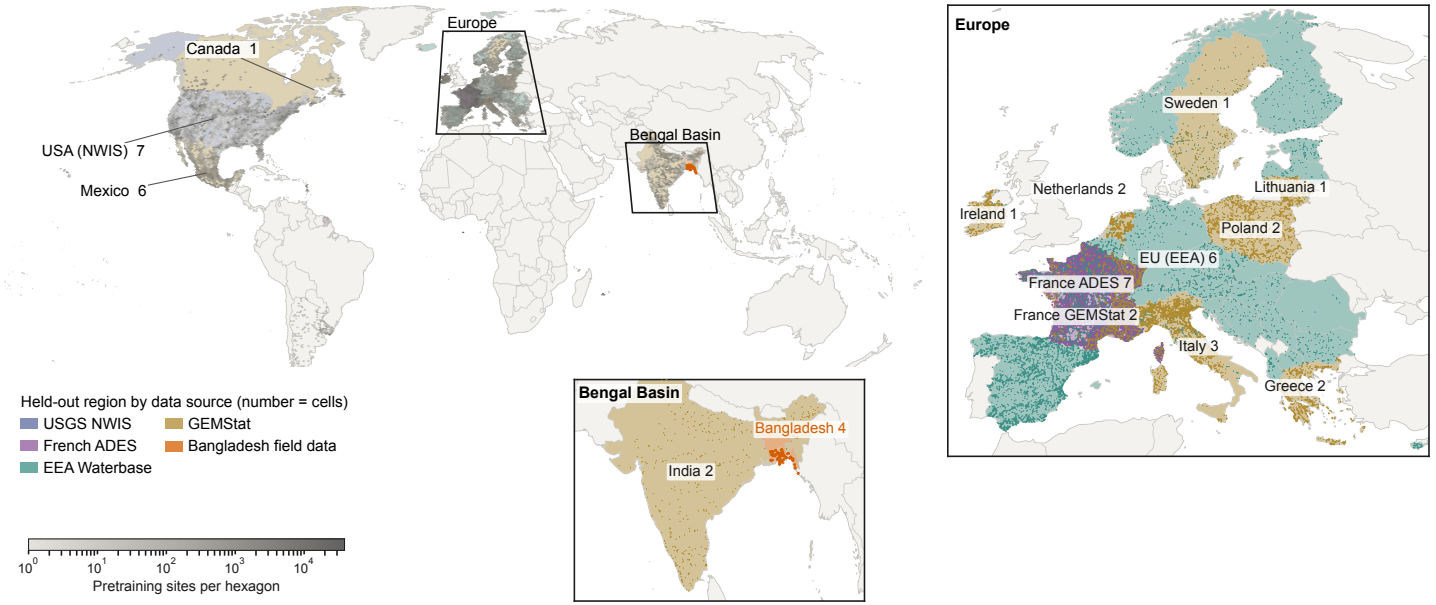
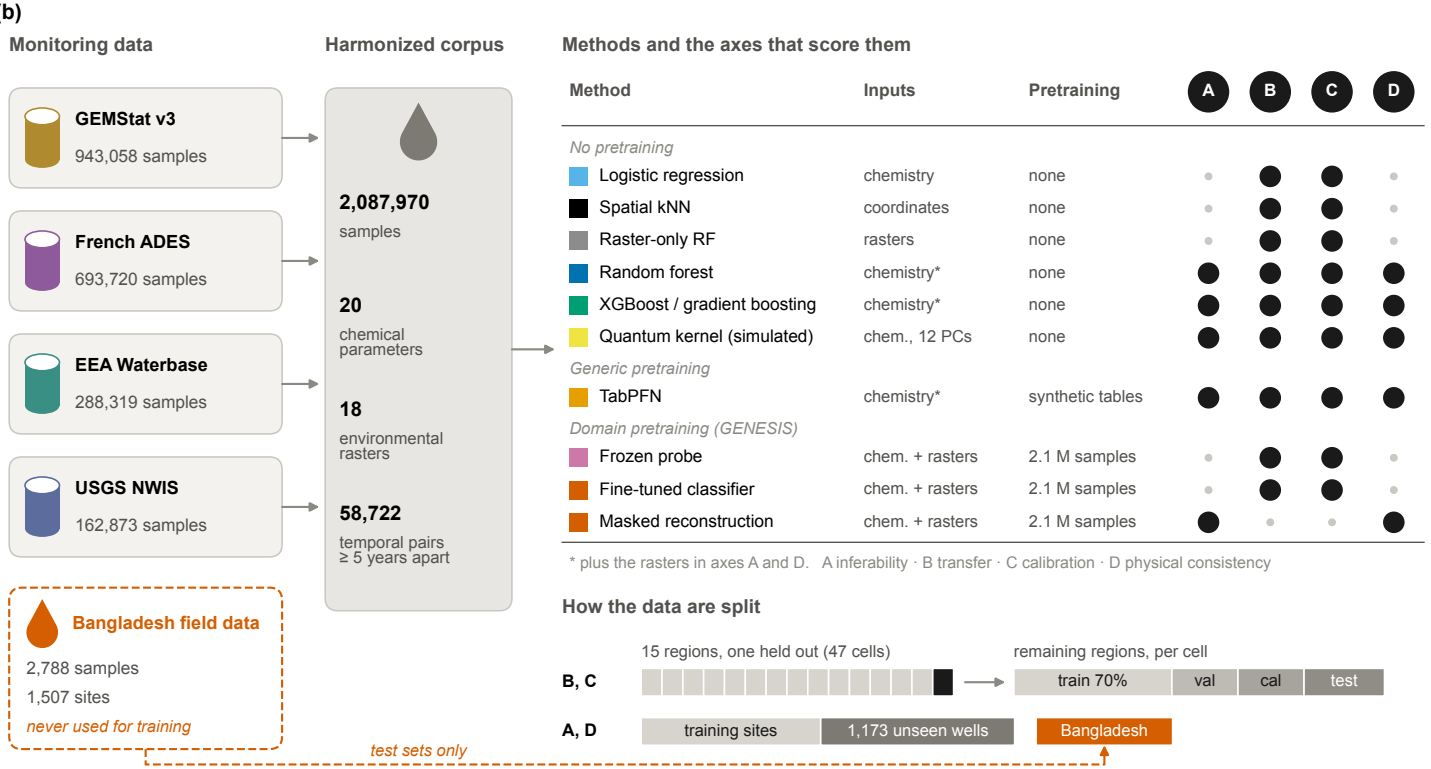
(a)


Europe
Canada 1
USA (NWIS) 7
Mexico 6
Bengal Basin
Europe
Sweden 1
Netherlands 2
Lithuania 1
Ireland 1
Poland 2
EU (EEA) 6
France ADES 7
France GEMStat 2
Italy 3
Greece 2
Bengal Basin
Bangladesh 4
India 2
Held-out region by data source (number = cells)
USGS NWIS
GEMStat
French ADES
Bangladesh field data
EEA Waterbase
Pretraining sites per hexagon

(b)


Monitoring data
Harmonized corpus
Methods and the axes that score them
GEMStat v3
943,058 samples
French ADES
693,720 samples
EEA Waterbase
288,319 samples
USGS NWIS
162,873 samples
2,087,970
samples
20
chemical parameters
18
environmental rasters
58,722
temporal pairs ≥ 5 years apart
Bangladesh field data
2,788 samples
1,507 sites
never used for training
test sets only
Method
Inputs
Pretraining
A
B
C
D
No pretraining
Logistic regression
chemistry
none
Spatial kNN
coordinates
none
Raster-only RF
rasters
none
Random forest
chemistry*
none
XGBoost / gradient boosting
chemistry*
none
Quantum kernel (simulated)
chem., 12 PCs
none
Generic pretraining
TabPFN
chemistry*
synthetic tables
Domain pretraining (GENESIS)
Frozen probe
chem. + rasters
2.1 M samples
Fine-tuned classifier
chem. + rasters
2.1 M samples
Masked reconstruction
chem. + rasters
2.1 M samples
* plus the rasters in axes A and D. A inferability · B transfer · C calibration · D physical consistency
How the data are split
15 regions, one held out (47 cells)
remaining regions, per cell
B, C
train 70%
val
cal
test
A, D
training sites
1,173 unseen wells
Bangladesh